DSAA3073: Theories in Data Science

Week 3: Agnostic PAC Learning, Uniform Convergence & Fundamental Theorem

Concise Learning Sheet

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Explorative projection

Opening Challenge

The Agnostic Learner's Dilemma

Last week we assumed the target concept $c \in \mathcal{H}$—the realizable case. But what if this assumption fails?

- Your training data is noisy: some labels are wrong
- Your hypothesis class is too simple: no $h \in \mathcal{H}$ perfectly fits the data
- You don't know which scenario you're in

Can we still learn? What guarantees can we provide? This week we tackle the **agnostic** setting where we make no assumptions about realizability, and discover the **Fundamental Theorem of Statistical Learning** that characterizes exactly when learning is possible.

Introduction

Last week we studied PAC learning in the **realizable** case, where we assumed there exists a perfect hypothesis $h^* \in \mathcal{H}$ with zero error. This week we remove this assumption and enter the **agnostic** setting—the real world where no perfect hypothesis may exist. The formal definition and full implications are developed in the next section.

Learning objectives: By the end of this week, you will:

1. Formalize agnostic PAC learning and understand **uniform convergence** (simultaneous convergence of empirical error to true error for all hypotheses)
2. Prove generalization bounds using **symmetrization techniques** (replacing unknown expectations with "ghost sample" empirical estimates)
3. Apply the Fundamental Theorem of Statistical Learning
4. Understand the **No Free Lunch theorem** (why unrestricted hypothesis classes cannot be learned) and its implications
5. Connect sample complexity to hypothesis class complexity

The learning journey this week: We'll start by formalizing the agnostic setting, then discover that ERM works if and only if **uniform convergence** holds. This leads us to ask: when does uniform convergence hold? The answer—finite VC dimension—is the Fundamental Theorem. We'll prove this using **symmetrization**, derive concrete sample complexity bounds, and see that restrictions on $\mathcal{H}$ are **necessary** (No Free Lunch). Finally, we'll see how to choose among multiple hypothesis classes (SRM).

Every concept builds on the last, forming a complete theory of when and how learning is possible.

Connection

From Week 2: We build on VC dimension from last week. Recall that $\mathrm{VCdim}(\mathcal{H})$ measures the capacity of $\mathcal{H}$ to shatter sets. This week, VC dimension becomes the key to understanding when agnostic learning is possible—see the Fundamental Theorem in Section 5.

Example: Recurring scenario: Medical diagnosis

Throughout this week, we'll follow this example: You're building a classifier to predict disease from symptoms using linear classifiers. The true relationship is nonlinear, so even the best linear classifier has 15% error. With 1000 noisy training examples, can you find a linear classifier with 17% error? Watch how each concept we develop applies to this scenario.

The Agnostic PAC Learning Framework

Motivation: Why Agnostic?

Thought Experiment

The Noisy World

You're building a spam classifier. Your training data comes from users who sometimes mislabel emails:

- Legitimate emails marked as spam (false positives)
- Spam emails marked as legitimate (false negatives)
- Inconsistent labeling: same email, different labels

Even if you had infinite data, no classifier could achieve zero error. The best you can do is match the performance of the optimal classifier in your hypothesis class.

Definition: Agnostic PAC Learning

A hypothesis class $\mathcal{H}$ is **agnostic PAC-learnable** if there exists an algorithm A and a sample-complexity function $m_{\mathcal{H}}(\varepsilon, \delta)$ (depending on error tolerance ε, confidence δ, and the complexity of $\mathcal{H}$) such that:

For all $\varepsilon > 0$, $\delta > 0$, and all distributions $\mathcal{D}$ over $\mathcal{X} \times \mathcal{Y}$ (joint distribution over inputs and labels):

If $m \geq m_{\mathcal{H}}(\varepsilon, \delta)$, then with probability at least $1 - \delta$ over $S \sim \mathcal{D}^m$ (the random training sample):

$$R(A(S)) \leq \inf_{h \in \mathcal{H}} R(h) + \varepsilon \tag{1}$$

where $R(h) = \mathbb{P}_{(x,y)\sim\mathcal{D}}[h(x) \neq y]$ is the **true risk** (probability of misclassification). For finite-VC classes, $m_{\mathcal{H}}$ is controlled by $\mathrm{VCdim}(\mathcal{H})$.

In words: The algorithm finds a hypothesis whose error is at most ε worse than the best possible hypothesis in $\mathcal{H}$. We're competing with the best in class, not trying to achieve zero error.

Technical note: If the infimum is not attained (no best hypothesis exists in $\mathcal{H}$), we use approximate ERM: find h_S with $\hat{R}_S(h_S) \leq \inf_{h \in \mathcal{H}} \hat{R}_S(h) + \frac{\varepsilon}{2}$. The analysis then gives $R(h_S) \leq \inf_{h \in \mathcal{H}} R(h) + \varepsilon$ as required.

Key differences from realizable case:

- **No assumption** that $\min_{h \in \mathcal{H}} R(h) = 0$ (there may be no perfect hypothesis)
- **Goal:** compete with best hypothesis in $\mathcal{H}$ (achieve $R(h) \leq \min_{h \in \mathcal{H}} R(h) + \varepsilon$), not achieve zero error
- **More realistic** but potentially harder (requires more samples)

Pause and Think

Question: In the realizable case, we required $R(A(S)) \leq \varepsilon$. In the agnostic case, we require $R(A(S)) \leq \min_{h \in \mathcal{H}} R(h) + \varepsilon$. Why is this weaker?

Hint: Consider what happens when $\min_{h \in \mathcal{H}} R(h) > 0$.

Answer:

> If the best hypothesis has error 0.3, the agnostic guarantee allows us to have error up to $0.3 + \varepsilon$, while the realizable guarantee would require error $\leq \varepsilon$. The agnostic setting acknowledges we can't beat the best in class.

✓ Checkpoint

Quick check: In the agnostic setting with $\min_{h \in \mathcal{H}} R(h) = 0.2$ and $\varepsilon = 0.05$, what error rate does PAC learning guarantee?

Answer: At most $0.2 + 0.05 = 0.25$ with probability $\geq 1 - \delta$. We compete with the best in class (0.2) plus tolerance ε.

Empirical Risk Minimization (ERM)

Empirical Risk Minimization (ERM) is the natural learning algorithm: choose the hypothesis that performs best on the training data.

Definition: Empirical Risk Minimization

Given training sample $S = \{(x_1, y_1), ..., (x_m, y_m)\}$, the **ERM algorithm** returns:

$$h_S = \operatorname*{argmin}_{h \in \mathcal{H}} \hat{R}_S(h) = \operatorname*{argmin}_{h \in \mathcal{H}} \frac{1}{m} \sum_{i=1}^{m} \mathbb{1}_{h(x_i) \neq y_i} \tag{2}$$

where $\hat{R}_S(h)$ is the **empirical risk** (fraction of training examples misclassified by h, defined in Week 2).

In words: ERM selects the hypothesis that makes the fewest mistakes on the training sample.

Intuition: If we can't identify the true best hypothesis (since we don't know the true distribution $\mathcal{D}$), at least find the one that performs best on our sample.

Wrong Path: Why Not Just Minimize Training Error?

"ERM always returns the hypothesis with lowest training error. Isn't that enough?"

Why this fails: Overfitting! A complex hypothesis class can memorize the training data but fail on new data.

Example: Nearest neighbor classifier achieves $\hat{R}_S(h) = 0$ on any dataset but may have terrible generalization.

The key question: When does ERM generalize? When is $R(h_S) \approx \min_{h \in \mathcal{H}} R(h)$?

Connection

Preview of uniform convergence: The answer is **uniform convergence**. We need $\hat{R}_S(h)$ to approximate $R(h)$ uniformly for **all** $h \in \mathcal{H}$ simultaneously. This is the bridge between training and test performance, and it's controlled by VC dimension.

Uniform Convergence

But wait—how do we **know** that ERM actually works? We defined ERM as the algorithm that minimizes training error, but we need $R(h_S) \approx \min_{h \in \mathcal{H}} R(h)$ to hold. What mathematical property guarantees this?

The answer is **uniform convergence**. This is the bridge between what we observe (training error) and what we care about (test error). Let's make this precise.

The Core Concept

Definition: Uniform Convergence

For i.i.d. samples and 0-1 loss in this section, we say that **uniform convergence** holds for hypothesis class $\mathcal{H}$ if:

$$\lim_{m \to \infty} \mathbb{P}_S \left[\sup_{h \in \mathcal{H}} |R(h) - \hat{R}_S(h)| > \varepsilon \right] = 0 \tag{3}$$

for all $\varepsilon > 0$ and all distributions $\mathcal{D}$ over $\mathcal{X} \times \mathcal{Y}$.

In words: As the sample size m grows, the probability that **any** hypothesis in $\mathcal{H}$ has empirical error far from its true error goes to zero. The convergence is **uniform**—it happens for all hypotheses simultaneously at the same rate.

Here $\sup_{h \in \mathcal{H}}$ denotes the **supremum** (least upper bound)—the worst-case deviation over all hypotheses. The key requirement: this worst-case deviation becomes small with high probability.

Finite-sample form: For any $\varepsilon, \delta > 0$, if $m \geq m_0(\varepsilon, \delta)$, then with probability at least $1 - \delta$:

$$\sup_{h \in \mathcal{H}} |R(h) - \hat{R}_S(h)| \leq \varepsilon \tag{4}$$

Interpretation: With enough samples, empirical risk $\hat{R}_S(h)$ approximates true risk $R(h)$ **uniformly** for all hypotheses in $\mathcal{H}$ simultaneously (not just for individual fixed hypotheses).

Key Takeaway

Uniform convergence is the bridge between training performance and test performance. If it holds, and if the relevant infima are attained (or if we use approximate ERM), then:

$$R(h_S) \leq \hat{R}_S(h_S) + \varepsilon \leq \hat{R}_S(h^*) + \varepsilon \leq R(h^*) + 2\varepsilon \tag{5}$$

where $h^* = \operatorname{argmin}_{h \in \mathcal{H}} R(h)$ and $h_S = \operatorname{argmin}_{h \in \mathcal{H}} \hat{R}_S(h)$.

Proof. Let $h^* = \operatorname{argmin}_{h \in \mathcal{H}} R(h)$ and $h_S = \operatorname{argmin}_{h \in \mathcal{H}} \hat{R}_S(h)$.

By uniform convergence, with high probability:

$$\forall h \in \mathcal{H} : |R(h) - \hat{R}_S(h)| \leq \varepsilon \tag{6}$$

Therefore:

$$\begin{aligned} R(h_S) &\leq \hat{R}_S(h_S) + \varepsilon && \text{(uniform convergence for } h_S\text{)} \\ &\leq \hat{R}_S(h^*) + \varepsilon && \text{(ERM minimizes empirical risk)} \\ &\leq R(h^*) + 2\varepsilon && \text{(uniform convergence for } h^*\text{)} \end{aligned} \tag{7}$$

■

Connection

Why "uniform"? We need the bound to hold for **all** hypotheses simultaneously, not just for a fixed hypothesis. This is much stronger than pointwise convergence and is essential for ERM to work.

🧠 Mental Model: Uniform vs. Pointwise Convergence

Pointwise: For each fixed h, as $m \to \infty$, $\hat{R}_S(h) \to R(h)$.

Uniform: For all $h \in \mathcal{H}$ simultaneously, $\sup_{h\in\mathcal{H}}|R(h) - \hat{R}_S(h)| \to 0$.

Think of pointwise as each hypothesis converging on its own timeline. Uniform means **all** hypotheses converge together, at a rate independent of which h we pick. This uniformity is what allows ERM to work.

✓ Checkpoint

Quick check: Why can't we just apply concentration inequalities to a single fixed hypothesis h to justify ERM?

Answer: Because ERM chooses h_S based on the data—it's data-dependent. We need bounds that hold uniformly for all $h \in \mathcal{H}$, including the one ERM selects.

Progress on backbone: Uniform convergence is the condition that makes ERM work. But when does it hold? This is the key question that the Fundamental Theorem will answer.

Finite Hypothesis Sets: Agnostic Bound

Theorem: Agnostic PAC Bound for Finite H

Let $\mathcal{H}$ be finite. For any $\varepsilon, \delta > 0$, if

$$m \geq \frac{1}{2\varepsilon^2}\left(\log|\mathcal{H}| + \log\left(\frac{2}{\delta}\right)\right) \tag{8}$$

then with probability at least $1 - \delta$ over $S \sim \mathcal{D}^m$:

$$\forall h \in \mathcal{H} : |R(h) - \hat{R}_S(h)| \leq \varepsilon \tag{9}$$

Applying this uniform-deviation bound with tolerance $\frac{\varepsilon}{2}$ gives the ERM excess-risk guarantee $R(h_S) \leq \inf_{h\in\mathcal{H}} R(h) + \varepsilon$ with sample complexity $m \geq \frac{2}{\varepsilon^2}\left(\log|\mathcal{H}| + \log\left(\frac{2}{\delta}\right)\right)$.

Proof (Guided Discovery):

Step 1: Fix a single hypothesis.

For fixed $h \in \mathcal{H}$, the empirical risk is:

$$\hat{R}_S(h) = \frac{1}{m}\sum_{i=1}^{m} \mathbb{1}_{h(x_i)\neq y_i} \tag{10}$$

Each indicator is an i.i.d. random variable in $[0, 1]$ with expectation $R(h)$.

By Hoeffding's inequality:

$$\mathbb{P}\left[|\hat{R}_S(\mathsf{h}) - R(h)| > \varepsilon\right] \leq 2\exp(-2m\varepsilon^2) \tag{11}$$

Step 2: Union bound over all hypotheses.

$$\begin{aligned}\mathbb{P}\left[\exists h \in \mathcal{H} : |R(h) - \hat{R}_S(\mathsf{h})| > \varepsilon\right] &\leq \sum_{h\in\mathcal{H}} \mathbb{P}\left[|R(h) - \hat{R}_S(\mathsf{h})| > \varepsilon\right] \\ &\leq |\mathcal{H}| \cdot 2\exp(-2m\varepsilon^2)\end{aligned} \tag{12}$$

Step 3: Solve for sample complexity.

Set $|\mathcal{H}| \cdot 2\exp(-2m\varepsilon^2) \leq \delta$:

$$\log|\mathcal{H}| + \log 2 - 2m\varepsilon^2 \leq \log\delta \tag{13}$$

$$m \geq \frac{1}{2\varepsilon^2}\left(\log|\mathcal{H}| + \log\left(\frac{2}{\delta}\right)\right) \tag{14}$$

Student Challenge

Exercise: Compare this bound to the realizable case bound from Week 2. Why does the agnostic case require $O(\frac{1}{\varepsilon^2})$ samples instead of $O(\frac{1}{\varepsilon})$?

Connection

From Week 2: In the realizable case, we had $m \geq \frac{1}{\varepsilon}(\log|\mathcal{H}| + \log(\frac{1}{\delta}))$ with $\frac{1}{\varepsilon}$ dependence. The agnostic case requires $\frac{1}{\varepsilon^2}$. For a detailed explanation of why agnostic learning needs the stronger $\frac{1}{\varepsilon^2}$ rate, see the key-takeaway box in the Fundamental Theorem section below.

✓ Checkpoint

Quick check: For $|\mathcal{H}| = 1000$, $\varepsilon = 0.1$, $\delta = 0.05$, how many samples does the agnostic bound require?

Answer: $m \geq \frac{1}{2\cdot 0.01}(\log 1000 + \log 40) \approx 50(6.9 + 3.7) = 530$ samples. Compare to realizable: $m \geq \frac{1}{0.1}(\log 1000 + \log 20) \approx 99$ samples—over 5× fewer!

The Fundamental Theorem of Statistical Learning

We've now seen that uniform convergence makes ERM succeed. But this raises deeper questions:

- **When** does uniform convergence hold?
- Is there a **simple condition** that characterizes learnability?
- What **exactly** controls sample complexity?

The Fundamental Theorem answers all three questions with a single condition: **finite VC dimension**. This is the crowning achievement of statistical learning theory—it completely characterizes when learning is possible.

Statement and Significance

Theorem: Fundamental Theorem of Statistical Learning

In the standard distribution-free binary 0-1 classification setting, for a suitably measurable hypothesis class $\mathcal{H}$ of functions from $\mathcal{X}$ to $\{0, 1\}$, the following are equivalent (each implies all the others):

1. $\mathcal{H}$ has the **uniform convergence property** (empirical risk converges to true risk uniformly for all $h \in \mathcal{H}$)
2. Any ERM rule is a **successful agnostic PAC learner** for $\mathcal{H}$
3. $\mathcal{H}$ is **agnostic PAC-learnable** (learnable without assuming a perfect hypothesis exists)
4. $\mathcal{H}$ is **PAC learnable**
5. Any ERM rule is a **successful PAC learner** for $\mathcal{H}$
6. $\mathcal{H}$ has **finite VC dimension** ($\mathrm{VCdim}(\mathcal{H}) < \infty$)

Moreover, if $\mathrm{VCdim}(\mathcal{H}) = d < \infty$, agnostic PAC learning has sample complexity roughly

$$m = O\left(\frac{d + \log(\frac{1}{\delta})}{\varepsilon^2}\right) \tag{15}$$

up to logarithmic factors depending on the theorem version, while realizable PAC learning has a faster $\frac{1}{\varepsilon}$ dependence (instead of $\frac{1}{\varepsilon^2}$).

In words: Within binary classification, having finite VC dimension is both necessary and sufficient for learning. If VC dimension is finite, the class is learnable (both in agnostic and realizable settings); if infinite, it's not learnable. As the theorem title suggests, this is the fundamental characterization of statistical learning.

Key Takeaway

Why $\frac{1}{\varepsilon^2}$ in agnostic vs. $\frac{1}{\varepsilon}$ in realizable?

The different sample complexity rates reflect fundamentally different problems:

Realizable case ($\frac{1}{\varepsilon}$ rate):

- We know some $h^* \in \mathcal{H}$ has zero error: $R(h^*) = 0$
- Task: **Detect** if $R(h) > \varepsilon$ (one-sided test)
- Hoeffding for detection: $\mathbb{P}\left[\hat{R}_S(h) \leq \frac{\varepsilon}{2} \mid R(h) \geq \varepsilon\right] \leq \exp\left(-m\frac{\varepsilon^2}{2}\right)$
- Need $m = O\left(\frac{1}{\varepsilon}\right)$ samples to reliably detect non-zero error

Agnostic case ($\frac{1}{\varepsilon^2}$ rate):

- No perfect hypothesis: $\min_{h \in \mathcal{H}} R(h)$ may be large
- Task: **Estimate** $R(h)$ within $\pm\varepsilon$ (two-sided estimation)
- Hoeffding for estimation: $\mathbb{P}\left[|\hat{R}_S(\mathsf{h}) - R(h)| > \varepsilon\right] \leq 2\exp(-2m\varepsilon^2)$
- Need $m = O\left(\frac{1}{\varepsilon^2}\right)$ samples for accurate estimation

The fundamental difference: Testing whether something equals zero requires fewer samples than estimating its actual value. This is why agnostic learning (which estimates) is harder than realizable learning (which only tests).

Significance: Within binary classification, this theorem **completely characterizes** when learning is possible. Every other property—uniform convergence, PAC learnability, sample complexity bounds—all reduce to this single combinatorial condition.

Key Takeaway

Translation table:

- **Uniform convergence:** empirical estimates are reliable for all h simultaneously
- **PAC learnability:** some algorithm succeeds with enough samples
- **VC dimension:** the combinatorial capacity of $\mathcal{H}$

The Fundamental Theorem says these viewpoints coincide in the standard VC setting.

Theorem Landscape

The Four Equivalent Statements

The theorem establishes a cycle: (4) Finite VC $\Rightarrow$ (1) Uniform convergence $\Rightarrow$ (2) Agnostic PAC $\Rightarrow$ (3) Realizable PAC $\Rightarrow$ (4) Finite VC

- (4)$\Rightarrow$(1): Combinatorics controls statistics (via Sauer's lemma)
- (1)$\Rightarrow$(2): Statistical reliability enables ERM to succeed
- (2)$\Rightarrow$(3): Agnostic learning includes realizable as special case
- (3)$\Rightarrow$(4): Necessity via No Free Lunch

Answering the backbone: Learning is possible $\iff$ finite VC dimension. This single condition unifies everything—uniform convergence, PAC learnability, and sample complexity all reduce to this one combinatorial property.

📊 **Confidence Check**

Self-assessment: Before continuing, ensure you can:

- Explain why uniform convergence is stronger than pointwise
- State the sample complexity: $O\left(\frac{d+\log(\frac{1}{\delta})}{\varepsilon^2}\right)$
- Articulate why finite VC dimension is both necessary and sufficient

Connection

To Week 4: The Fundamental Theorem tells us **when** learning is possible. Next week: **which** hypothesis class to use when we have choices (model selection, regularization, cross-validation).

Remark

Two-sided vs. one-sided bounds: Uniform convergence naturally gives two-sided bounds: $|R(h) - \hat{R}_S(h)| \leq \varepsilon$ (bounds deviations in both directions). For ERM analysis, we only need the one-sided bound $R(h) \leq \hat{R}_S(h) + \varepsilon$ (bounding how much true risk can exceed empirical risk). Both follow from the same uniform convergence property, but the one-sided version is sufficient for proving PAC learnability.

Proof Sketch: The Equivalence Chain

We prove the equivalences in a cycle: (4) $\Rightarrow$ (1) $\Rightarrow$ (2) $\Rightarrow$ (3) $\Rightarrow$ (4).

Part 1: Finite VC dimension implies uniform convergence

💪 **Productive Struggle**

Before reading the proof: How would you prove that finite VC dimension implies uniform convergence? The challenge: $\mathcal{H}$ might be infinite, so we can't use union bounds directly. But VC dimension controls the growth function...

Hint: The key insight is **symmetrization**—replace the unknown true risk with empirical risk on an independent "ghost sample."

Lemma: VC Dimension and Uniform Convergence

If $\mathrm{VCdim}(\mathcal{H}) = d < \infty$, then for any $\varepsilon, \delta > 0$, if

$$m \geq C \frac{d \log\left(\frac{1}{\varepsilon}\right) + \log\left(\frac{1}{\delta}\right)}{\varepsilon^2} \tag{16}$$

for some universal constant C, then with probability $\geq 1 - \delta$:

$$\forall h \in \mathcal{H} : |R(h) - \hat{R}_S(h)| \leq \varepsilon \tag{17}$$

Note on one-sided vs two-sided bounds: This is a two-sided bound (absolute value). For ERM analysis, we typically need only the one-sided bound $R(h) \leq \hat{R}_S(h) + \varepsilon$, but uniform convergence naturally gives the stronger two-sided guarantee. Both hold with the same sample complexity.

Proof idea: The standard proof does not directly union-bound over $\Pi_{\mathcal{H}}(m)$ on the same random sample. Instead:

Step 1: Symmetrize.

Introduce an independent ghost sample S' of size m. Large deviation between $R(h)$ and $\hat{R}_S(\mathtt{h})$ implies, with controlled constants, a large deviation between empirical errors on S and S'.

Step 2: Condition on the combined sample.

Condition on the $2m$ points in $S \cup S'$. On these fixed points, hypotheses matter only through their label patterns.

Step 3: Union bound over effective behaviors.

By Sauer's lemma, the number of distinct dichotomies on the combined sample is at most

$$\Pi_{\mathcal{H}}(2m) \leq \left(\frac{2em}{d}\right)^d \tag{18}$$

for $1 \leq d \leq 2m$.

Step 4: Apply concentration and solve for m.

A union bound over these label patterns, followed by Hoeffding-type concentration, gives

$$m \geq C \frac{d \log\left(\frac{1}{\varepsilon}\right) + \log\left(\frac{1}{\delta}\right)}{\varepsilon^2} \tag{19}$$

for a universal constant C.

Pause and Think

Question: Why does the proof use a ghost sample of size m and then $\Pi_{\mathcal{H}}(2m)$?

Hint: The unknown true risk is replaced by empirical behavior on an independent sample. After conditioning, both samples together contain $2m$ points.

Answer:

> We only need to count distinct label patterns on those $2m$ points, which is controlled by Sauer's lemma.

Connection

From Week 2: Sauer's lemma states $\Pi_{\mathcal{H}}(m) \leq \left(\frac{em}{d}\right)^d$ when $\mathrm{VCdim}(\mathcal{H}) = d < \infty$. This polynomial bound (not exponential 2^m) makes union bounds work for infinite classes. The growth function counts distinct label patterns on m points—exactly what we need here!

🪜 **Scaffolded Derivation (7 steps)**

Step-by-step: Why $\Pi_{\mathcal{H}}(2m)$ appears

1. Start: $\mathbb{P}_S\left[\sup_{h\in\mathcal{H}}|R(h) - \hat{R}_S(\mathsf{h})| > \varepsilon\right]$
2. Introduce ghost S': Replace $R(h)$ with $\frac{1}{m}\sum_{i=1}^m \mathbb{1}_{h(z_i')\neq y_i'}$
3. Now comparing two empirical quantities: $|\hat{R}_S(\mathsf{h}) - \hat{R}_{S'}(h)|$
4. Condition on combined sample $S \cup S'$ (total $2m$ points)
5. On these $2m$ points, distinct hypotheses $\leq \Pi_{\mathcal{H}}(2m)$
6. Union bound over $\Pi_{\mathcal{H}}(2m) \leq \left(\frac{2em}{d}\right)^d$ patterns
7. Result: $m = O\left(\frac{d\log(\frac{1}{\varepsilon}) + \log(\frac{1}{\delta})}{\varepsilon^2}\right)$

Part 2: Uniform convergence implies agnostic PAC learnability

This follows directly from our earlier proof that uniform convergence implies ERM succeeds.

Part 3: Agnostic PAC learnability implies realizable PAC learnability

Trivial: the realizable case is a special case of the agnostic case (when $\min_{h\in\mathcal{H}} R(h) = 0$).

Part 4: Realizable PAC learnability implies finite VC dimension

💪 **Productive Struggle**

The No Free Lunch construction: To prove this, show that infinite VC dimension makes learning impossible. If $\mathrm{VCdim}(\mathcal{H}) = \infty$, then for any sample size m, there exists a set of size $2m$ that $\mathcal{H}$ shatters. We can construct a distribution where the learner sees only half the points, and labels on the unseen half are random and independent of what the learner observed. Why does this make learning impossible?

Lemma: No Free Lunch for Infinite VC Dimension

If $\text{VCdim}(\mathcal{H}) = \infty$, then $\mathcal{H}$ is not PAC-learnable.

Proof (by contradiction):

Assume $\text{VCdim}(\mathcal{H}) = \infty$ but $\mathcal{H}$ is PAC-learnable by algorithm A.

Since VC dimension is infinite, for any m, there exists a set $C = \{x_1, ..., x_{2m}\}$ of size $2m$ that is shattered by $\mathcal{H}$.

Consider the following adversarial strategy:

1. Nature chooses distribution $\mathcal{D}$ uniform over C
2. Nature chooses a random labeling $c : C \to \{0, 1\}$
3. Algorithm A receives sample S of size m from $\mathcal{D}$ labeled by c

Pause and Think

Question: How many points from C are likely to remain unseen?

Hint: A fixed point is unseen with probability $\left(1 - \frac{1}{2m}\right)^m \approx e^{-\frac{1}{2}}$.

Answer:

> So a constant fraction of the $2m$ points remain unseen in expectation. On those unseen points, the random labels are independent of the sample the algorithm observed.

Key observation: Since C is shattered, for any labeling of the m unseen points, there exists $h \in \mathcal{H}$ consistent with the entire sample S and that labeling.

Why this matters: Consider what happens when the learner sees sample S drawn from the $2m$ points:

- The learner observes roughly m labeled points from C (those that fell in S)
- Roughly m points from C remain unseen (with labels determined by the random target)
- On the seen points, the learner can find a hypothesis with zero empirical error (since C is shattered)
- But there are 2^m possible labelings of the unseen m points, and each is equally likely given what the learner observed

The contradiction: Fix any hypothesis h the learner might output. Since all 2^m labelings of the unseen points are consistent with the training data:

- For half of these labelings (chosen uniformly at random), hypothesis h will make errors on at least half the unseen points
- The unseen points constitute a constant fraction ($\approx e^{-\frac{1}{2}} \approx 0.37$) of all $2m$ points
- Therefore, h has expected true error of at least $0.25 \cdot 0.37 > 0.09$ on a constant fraction of the possible target concepts

This holds for **any** hypothesis the learner outputs, regardless of the training sample size m. But PAC learnability requires that for any $\varepsilon > 0$, we can make $R(A(S)) \leq \varepsilon$ with high probability by choosing m large enough. This is impossible when the expected error is bounded below by a constant. Contradiction!

Connection

The No Free Lunch Theorem: This proof shows that without restrictions on $\mathcal{H}$, learning is impossible. Infinite VC dimension means the hypothesis class is "too rich" to learn from finite samples.

Note on rigor: This proof sketch is informal—a fully rigorous proof requires concentration inequalities to bound the number of unseen points and their contribution to the error. See Mohri et al. Theorem 3.21 or Shalev-Shwartz & Ben-David Theorem 6.7 for complete details.

✓ **Checkpoint**

Quick check: Why doesn't No Free Lunch contradict universal approximators like neural networks?

Answer: Neural networks with fixed architecture have **finite** VC dimension (though very large). No Free Lunch applies only to **infinite** VC dimension. The theorem says you must restrict your hypothesis class—which we do by fixing architecture.

Rademacher Complexity: A Technical Tool

Before diving into symmetrization, we need to introduce **Rademacher complexity**—a key technical tool that measures how well a function class can fit random noise. While VC dimension gives a combinatorial measure of complexity, Rademacher complexity provides a probabilistic measure that's crucial for proving generalization bounds.

Definition: Rademacher Complexity

Let $\mathcal{G}$ be a class of functions $\mathcal{Z} \to \mathbb{R}$ and $S = \{z_1, ..., z_m\}$ a fixed sample.

Empirical Rademacher complexity:

$$\hat{\mathcal{R}}_S(\mathcal{G}) = \mathbb{E}_\sigma \left[\sup_{g \in \mathcal{G}} \frac{1}{m} \sum_{i=1}^{m} \sigma_i g(z_i) \right] \tag{20}$$

where $\sigma = (\sigma_1, ..., \sigma_m)$ and each σ_i is an independent random variable uniform on $\{-1, +1\}$ (called **Rademacher variables**).

Expected Rademacher complexity:

$$\mathcal{R}_m(\mathcal{G}) = \mathbb{E}_S \left[\hat{\mathcal{R}}_S(\mathcal{G}) \right] \tag{21}$$

where the expectation is over random samples $S \sim \mathcal{D}^m$.

In words:

- $\hat{\mathcal{R}}_S(\mathcal{G})$ measures how well functions in $\mathcal{G}$ can correlate with random ± 1 noise on a **specific** sample S
- $\mathcal{R}_m(\mathcal{G})$ is the expected correlation over all possible samples of size m
- The Rademacher variables σ_i introduce randomness independent of the data

Intuition: If a function class can fit random noise well (high Rademacher complexity), it's expressive enough to overfit. Low Rademacher complexity means the class is restricted enough to prevent fitting noise—a good property for generalization.

Why "Rademacher"? The random signs $\sigma_i \in \{-1, +1\}$ are called Rademacher random variables (introduced by Hans Rademacher in 1922). They're the simplest non-trivial symmetric random variables.

Example: Rademacher Complexity: Intuition

Consider fitting random labels $\sigma_i \in \{-1, +1\}$ on m points:

- **Finite class:** If $|\mathcal{G}| = N$, then $\mathcal{R}_m(\mathcal{G}) \leq \sqrt{\frac{\log N}{m}}$
- **Linear functions in $\mathbb{R}^d$:** $\mathcal{R}_m(\mathcal{G}) = O\left(\sqrt{\frac{d}{m}}\right)$
- **All functions:** $\mathcal{R}_m(\mathcal{G}) = 1$ (can perfectly correlate with any noise)

Notice: more complex classes have higher Rademacher complexity.

Connection

Relationship to VC dimension: For 0-1 loss classes with $\mathrm{VCdim}(\mathcal{H}) = d$:

$$\mathcal{R}_m(\mathcal{G}) = O\left(\sqrt{\frac{d}{m}}\right) \tag{22}$$

where $\mathcal{G} = \left\{(x, y) \mapsto \mathbb{1}_{h(x)\neq y} : h \in \mathcal{H}\right\}$. This connection (via Massart's lemma and Sauer's lemma) is how VC dimension controls generalization through Rademacher complexity.

Key Takeaway

Rademacher complexity bridges combinatorial complexity (VC dimension) and statistical generalization. It's the technical tool that makes symmetrization work.

Symmetrization and Ghost Samples

The Fundamental Theorem tells us that finite VC dimension implies uniform convergence. But **how** do we prove this? The challenge is that $\mathcal{H}$ might be infinite, so we can't use union bounds directly.

The key technique is **symmetrization**—a beautiful proof method that replaces unknown expectations (true risk $R(h)$) with computable empirical averages on an independent "ghost sample." This transforms a problem about unknown distributions into one we can analyze combinatorially.

Understanding symmetrization isn't just about this proof—it's a fundamental technique you'll see throughout learning theory, information theory, and probability.

The Symmetrization Technique

One of the most powerful proof techniques in learning theory is **symmetrization**: replacing an unknown expectation (the true risk $R(h) = \mathbb{E}[\texttt{loss}]$) with an empirical average on an independent "ghost sample."

Definition: Ghost Sample

Given a sample $S = \{z_1, ..., z_m\}$ (where each $z_i = (x_i, y_i)$ is a labeled example), a **ghost sample** is an independent sample $S' = \{z'_1, ..., z'_m\}$ drawn from the same distribution $\mathcal{D}$.

In words: A ghost sample is a second, independent training set of the same size, drawn from the same distribution. We use it as a theoretical device to replace unknown expectations with computable empirical averages.

Key property: Both S and S' have the same distribution, so $\mathbb{E}_{S'}[g(z')] = \mathbb{E}_S[g(z)]$ for any function g.

Key idea: We can't compute $\mathbb{E}[g(z)]$ directly (since we don't know the distribution $\mathcal{D}$), but we can approximate it with $\frac{1}{m}\sum_{i=1}^{m} g(z'_i)$ using an independent ghost sample. This converts a problem involving unknown distributions into one involving only empirical quantities.

Theorem: Symmetrization Lemma (One-Sided Version)

Assumptions:

- Functions $g : \mathcal{Z} \to [0,1]$ are bounded (take values in $[0,1]$)
- Samples z_i, z_i' are i.i.d. from distribution $\mathcal{D}$

Statement: Let $\mathcal{G}$ be a class of functions $\mathcal{Z} \to [0,1]$, and let z_i be i.i.d. from distribution $\mathcal{D}$. Then:

$$\mathbb{E}_S\left[\sup_{g\in\mathcal{G}}\left(\mathbb{E}_Z[g(Z)] - \frac{1}{m}\sum_{i=1}^m g(z_i)\right)\right] \le 2\mathbb{E}_{S,S'}\left[\sup_{g\in\mathcal{G}}\frac{1}{m}\sum_{i=1}^m (g(z_i') - g(z_i))\right] \tag{23}$$

where $S = \{z_1, ..., z_m\}$ and $S' = \{z_1', ..., z_m'\}$ are independent samples from $\mathcal{D}$.

In words: The expected worst-case **one-sided** deviation between the true expectation and the sample average (left side) is bounded by twice the expected worst-case difference between two independent sample averages (right side). This replaces an unknown expectation with an empirical quantity we can analyze.

Key benefit: The right side involves only empirical quantities (comparing two samples), which we can bound using concentration inequalities and combinatorial arguments (like Sauer's lemma).

Important: This is the **one-sided** version (no absolute value on the left). There is also a two-sided version that bounds $\mathbb{E}\left[\sup|\mathbb{E}[g] - \frac{1}{m}\sum g(z_i)|\right]$—interestingly, both versions have the **same factor 2** on the right-hand side, because the factor 2 comes from the triangle inequality $|a - b| \le |a| + |b|$ applied to split the Rademacher terms (see Step 4), not from handling absolute values on the left. For ERM analysis we only need this one-sided version, which is sufficient to prove PAC learnability.

Tightness note: The one-sided bound may appear loose (why pay factor 2 if we only bound one direction?), but this is unavoidable with the symmetrization technique. More refined analyses using localization or PAC-Bayes can sometimes improve constants, but symmetrization remains the standard approach due to its simplicity and generality.

Proof (Guided Discovery):

Step 1: Introduce ghost sample.

$$\begin{aligned}\mathbb{E}_S\left[\sup_{g\in\mathcal{G}}\left(\mathbb{E}_Z[g(Z)] - \frac{1}{m}\sum_{i=1}^m g(z_i)\right)\right] &= \mathbb{E}_S\left[\sup_{g\in\mathcal{G}}\left(\mathbb{E}_{z'}[g(z')] - \frac{1}{m}\sum_{i=1}^m g(z_i)\right)\right] \\ &= \mathbb{E}_S\left[\sup_{g\in\mathcal{G}}\mathbb{E}_{S'}\left[\left(\frac{1}{m}\sum_{i=1}^m g(z_i')\right) - \left(\frac{1}{m}\sum_{i=1}^m g(z_i)\right)\right]\right]\end{aligned} \tag{24}$$

Step 2: Move supremum outside expectation.

By monotonicity of expectation and the inequality $\sup_g \mathbb{E}_{S'}[X_g] \le \mathbb{E}_{S'}[\sup_g X_g]$:

$$\mathbb{E}_S\left[\sup_{g\in\mathcal{G}}\mathbb{E}_{S'}[...]\right] \le \mathbb{E}_{S,S'}\left[\sup_{g\in\mathcal{G}}\left(\frac{1}{m}\sum_{i=1}^m (g(z_i') - g(z_i))\right)\right] \tag{25}$$

Step 3: Introduce Rademacher variables.

Let $\sigma = (\sigma_1, ..., \sigma_m)$ where σ_i are i.i.d. uniform on $\{-1, +1\}$.

Pause and Think

Question: Why is $(g(z_i') - g(z_i))$ symmetric in distribution to $\sigma_i(g(z_i') - g(z_i))$?

Hint: Both z_i and z_i' are i.i.d. from $\mathcal{D}$, so swapping them doesn't change the distribution.

Answer:

> The difference $g(z_i') - g(z_i)$ has the same distribution as $g(z_i) - g(z_i') = -(g(z_i') - g(z_i))$. Multiplying by random sign σ_i preserves this symmetry.

By symmetry:

$$\mathbb{E}_{S,S'}\left[\sup_{g\in\mathcal{G}} \frac{1}{m}\sum_{i=1}^{m}(g(z_i') - g(z_i))\right] = \mathbb{E}_{\sigma,S,S'}\left[\sup_{g\in\mathcal{G}} \frac{1}{m}\sum_{i=1}^{m}\sigma_i(g(z_i') - g(z_i))\right] \tag{26}$$

Step 4: Triangle inequality.

The key is to bound the supremum of a sum. For any fixed σ, S, S', let g^* be the function achieving the supremum. Then:

$$\sup_{g\in\mathcal{G}} \frac{1}{m}\sum_{i=1}^{m}\sigma_i(g(z_i') - g(z_i)) = \frac{1}{m}\sum_{i=1}^{m}\sigma_i(g^*(z_i') - g^*(z_i)) \tag{27}$$

Now apply the triangle inequality for absolute values. Note that:

$$\begin{aligned} |\sigma_i(g^*(z_i') - g^*(z_i))| &= |\sigma_i g^*(z_i') - \sigma_i g^*(z_i)| \\ &\leq |\sigma_i g^*(z_i') - 0| + |0 - \sigma_i g^*(z_i)| \\ &= |\sigma_i g^*(z_i')| + |\sigma_i g^*(z_i)| \end{aligned} \tag{28}$$

Here, $c = 0$ is the intermediate point in the triangle inequality $|a - b| \leq |a - c| + |c - b|$.

Summing over i and taking supremum:

$$\begin{aligned} \sup_{g\in\mathcal{G}} \left|\frac{1}{m}\sum_{i=1}^{m}\sigma_i(g(z_i') - g(z_i))\right| &\leq \sup_{g\in\mathcal{G}} \frac{1}{m}\sum_{i=1}^{m}|\sigma_i(g(z_i') - g(z_i))| \\ &\leq \sup_{g\in\mathcal{G}} \frac{1}{m}\sum_{i=1}^{m}(|\sigma_i g(z_i')| + |\sigma_i g(z_i)|) \\ &= \sup_{g\in\mathcal{G}} \frac{1}{m}\sum_{i=1}^{m}|\sigma_i g(z_i')| + \sup_{g\in\mathcal{G}} \frac{1}{m}\sum_{i=1}^{m}|\sigma_i g(z_i)| \end{aligned} \tag{29}$$

Since $g : \mathcal{Z} \to [0, 1]$ are non-negative and $\sigma_i \in \{-1, +1\}$, we have $|\sigma_i g(z)| = g(z)$. Therefore:

$$\begin{aligned} \mathbb{E}_{\sigma,S,S'}\left[\sup_{g\in\mathcal{G}} \frac{1}{m}\sum_{i=1}^{m}\sigma_i(g(z_i') - g(z_i))\right] &\leq \mathbb{E}_{\sigma,S'}\left[\sup_{g\in\mathcal{G}} \frac{1}{m}\sum_{i=1}^{m}\sigma_i g(z_i')\right] \\ &\quad + \mathbb{E}_{\sigma,S}\left[\sup_{g\in\mathcal{G}} \frac{1}{m}\sum_{i=1}^{m}\sigma_i g(z_i)\right] \end{aligned} \tag{30}$$

Since S and S' have the same distribution:

$$\ldots = 2\mathbb{E}_{\sigma,S}\left[\sup_{g\in\mathcal{G}} \frac{1}{m}\sum_{i=1}^{m} \sigma_i g(z_i)\right] \tag{31}$$

This final term is exactly $2\hat{\mathcal{R}}_S(\mathcal{G})$, the empirical Rademacher complexity.

Key Takeaway

Symmetrization converts a problem involving unknown expectations into one involving only empirical quantities. The price is a factor of 2 and the introduction of Rademacher variables.

Remark

Why does one-sided symmetrization still have factor 2?

You might wonder: "If we only bound one direction (no absolute value), why do we pay the same factor of 2 as the two-sided version?"

Answer: The factor 2 comes from the **triangle inequality** applied to Rademacher terms in Step 4:

$$\sigma_i(g(z_i') - g(z_i)) = \sigma_i g(z_i') - \sigma_i g(z_i) \tag{32}$$

We bound:

$$|\sigma_i g(z_i') - \sigma_i g(z_i)| \leq |\sigma_i g(z_i')| + |\sigma_i g(z_i)| \tag{33}$$

This splitting introduces factor 2 regardless of whether the original LHS has absolute values —it's a consequence of how we handle the ghost sample difference.

Is this loose? Yes, potentially! The symmetrization technique is not optimal in terms of constants. More refined approaches exist:

- **Localization techniques** can reduce constants in some cases
- **PAC-Bayes bounds** use different proof techniques with better constants
- **Direct Rademacher analysis** without ghost samples can be tighter

However, symmetrization remains the standard pedagogical approach because:

1. Conceptually clear (replace unknown with ghost sample)
2. Works for general function classes
3. Leads directly to VC dimension characterization
4. The factor 2 is absorbed by other constants in practice (the $O(\cdot)$ notation)

The takeaway: Factor 2 is the "tax" we pay for the symmetrization technique, not a fundamental limit of learning theory.

Wrong Path: Common Confusion: Where Does the Factor 2 Go?

The Question: "The Symmetrization Lemma has a factor of 2. But when we apply it to get the generalization bound (next theorem), we still have factor 2 in front of the square root. Shouldn't it be 4 after combining constants?"

The Answer: The factor 2 in the Symmetrization Lemma is **correctly preserved** in the final bound! Here's why:

The Full Chain:

1. Symmetrization: $\mathbb{E}\left[\sup\left(R(h) - \hat{R}_S(\mathsf{h})\right)\right] \leq 2 \cdot \hat{\mathcal{R}}_m(\mathcal{H}) \quad \leftarrow$ factor **2**
2. Massart's Lemma: $\hat{\mathcal{R}}_m(\mathcal{H}) \leq \sqrt{\frac{\log|\mathcal{H}|_S}{m}} \quad \leftarrow$ no extra factor
3. Sauer's Lemma: $\log|\mathcal{H}|_S \leq d \log\left(\frac{em}{d}\right) \quad \leftarrow$ no extra factor
4. Combined: $\mathbb{E}\left[\sup\left(R(h) - \hat{R}_S(\mathsf{h})\right)\right] \leq 2\sqrt{\frac{d\log(em/d)}{m}} \quad \leftarrow$ factor **2** preserved
5. McDiarmid: Add $\sqrt{\frac{\log(2/\delta)}{2m}}$ for high-probability bound

Final bound: $R(h) \leq \hat{R}_S(\mathsf{h}) + 2\sqrt{\frac{2d\log\left(\frac{2em}{d}\right)}{m}} + \sqrt{\frac{\log(2/\delta)}{2m}}$

The factor **2** in front comes directly from symmetrization step 1. The factor **2** inside the log comes from Sauer's lemma refinements. These are different 2′s!

Key insight: We're bounding $\sup\left(R(h) - \hat{R}_S(\mathsf{h})\right)$ (one-sided, no absolute value), which is exactly what ERM needs. The factor 2 from symmetrization is the **only** factor 2 in the main term, and it appears explicitly in the final bound.

Connection

Connecting to VC dimension via Massart's Lemma: The Rademacher complexity $\hat{\mathcal{R}}_S(\mathcal{G})$ can be bounded using the growth function. Massart's lemma states that for functions taking values in $\{0, 1\}$:

$$\hat{\mathcal{R}}_S(\mathcal{G}) \leq \sqrt{\frac{\log|\mathcal{G}|_S}{m}} \tag{34}$$

where $|\mathcal{G}|_S$ is the number of distinct functions in $\mathcal{G}$ on sample S. By Sauer's lemma, $|\mathcal{G}|_S \leq \Pi_{\mathcal{H}}(m) \leq \left(\frac{em}{d}\right)^d$, giving $\hat{\mathcal{R}}_S(\mathcal{G}) = O\left(\sqrt{\frac{d\log m}{m}}\right)$. This is how VC dimension controls generalization through Rademacher complexity. See Mohri et al. Theorem 3.7 for the full proof.

🔨 **Construct Your Own**

Build a toy symmetrization: Consider hypothesis h with $R(h) = 0.3$ and sample S of size $m = 100$.

1. What's $\mathbb{E}\left[\hat{R}_S(\mathsf{h})\right]$?
2. Introduce ghost sample S'. What's $\mathbb{E}\left[\hat{R}_{S'}(h)\right]$?
3. What's $\mathbb{E}\left[\hat{R}_S(h) - \hat{R}_{S'}(h)\right]$?
4. How does this relate to $R(h) - \hat{R}_S(h)$?

This reveals why replacing $R(h)$ with ghost sample average preserves the expectation structure.

Connection

To Week 5: Symmetrization leads to **Rademacher complexity**, a more refined measure than VC dimension. Rademacher gives data-dependent bounds that can be much tighter in practice.

Application to Generalization Bounds

Using symmetrization, we can derive generalization bounds without explicitly computing expectations.

Remark

Roadmap to the Generalization Bound:

We now apply the Symmetrization Lemma to derive a concrete generalization bound. The steps are:

1. **Symmetrization** (proved above): Replace $R(h)$ with comparison between two samples $\rightarrow$ factor **2**
2. **Rademacher bound via Massart**: Bound empirical Rademacher using $|\mathcal{H}|_S$
3. **Growth function via Sauer**: Bound $|\mathcal{H}|_S$ using VC dimension d
4. **Concentration via McDiarmid**: Convert expectation to high-probability statement

The factor of 2 from step 1 will appear explicitly in the final bound—watch for it!

Theorem: Generalization Bound via Symmetrization

In binary 0-1 classification with i.i.d. samples, suppose $\mathrm{VCdim}(\mathcal{H}) = d$ and $1 \leq d \leq m$. For any $\delta > 0$, with probability $\geq 1 - \delta$:

$$\forall h \in \mathcal{H} : R(h) \leq \hat{R}_S(h) + 2\sqrt{\frac{2d\log\left(\frac{2em}{d}\right)}{m}} + \sqrt{\frac{\log\left(\frac{2}{\delta}\right)}{2m}} \tag{35}$$

Proof sketch:

Step 1: Symmetrization → Rademacher complexity

Apply the Symmetrization Lemma to bound the expected supremum:

$$\mathbb{E}\left[\sup_{h\in\mathcal{H}}\left(R(h)-\hat{R}_S(\mathsf{h})\right)\right] \leq 2\hat{\mathcal{R}}_m(\mathcal{H}) \tag{36}$$

This converts unknown risk $R(h)$ into empirical Rademacher complexity (factor 2 from triangle inequality).

Step 2: Massart's Lemma → Growth function

Bound the Rademacher complexity using the number of distinct functions on the sample:

$$\hat{\mathcal{R}}_m(\mathcal{H}) \leq \sqrt{\frac{\log|\mathcal{H}|_S}{m}} \tag{37}$$

Step 3: Sauer's Lemma → VC dimension

Bound the growth function using VC dimension:

$$\log|\mathcal{H}|_S \leq d\log\left(\frac{em}{d}\right) \tag{38}$$

Combining Steps 1-3 gives the **expectation** bound:

$$\mathbb{E}\left[\sup_{h\in\mathcal{H}}\left(R(h)-\hat{R}_S(\mathsf{h})\right)\right] \leq 2\sqrt{\frac{d\log\left(\frac{em}{d}\right)}{m}} \tag{39}$$

Step 4: McDiarmid's Inequality → High-probability bound

The expectation bound tells us the **average** behavior, but PAC learning requires **high-probability** guarantees. McDiarmid's inequality provides concentration: if a function has bounded differences (changing one sample point changes the supremum by at most $\frac{1}{m}$), then with probability $\geq 1-\delta$:

$$\sup_{h\in\mathcal{H}}\left(R(h)-\hat{R}_S(\mathsf{h})\right) \leq \mathbb{E}\left[\sup_{h\in\mathcal{H}}\left(R(h)-\hat{R}_S(\mathsf{h})\right)\right] + \sqrt{\frac{\log\left(\frac{1}{\delta}\right)}{2m}} \tag{40}$$

Substituting the expectation bound from Step 3:

$$\sup_{h\in\mathcal{H}}\left(R(h)-\hat{R}_S(\mathsf{h})\right) \leq 2\sqrt{\frac{d\log\left(\frac{em}{d}\right)}{m}} + \sqrt{\frac{\log\left(\frac{1}{\delta}\right)}{2m}} \tag{41}$$

Final result: With probability $\geq 1-\delta$, for all $h\in\mathcal{H}$:

$$R(h) \leq \hat{R}_S(\mathsf{h}) + 2\sqrt{\frac{2d\log\left(\frac{2em}{d}\right)}{m}} + \sqrt{\frac{\log\left(\frac{2}{\delta}\right)}{2m}} \tag{42}$$

Key observations:

- Factor 2 in the main term comes from symmetrization (Step 1)
- The additive confidence term $\sqrt{\frac{\log(\frac{2}{\delta})}{2m}}$ comes from McDiarmid (Step 4)
- This converts "in expectation" to "with high probability"

Key Takeaway

McDiarmid's Inequality

Statement: Let $X_1, ..., X_m$ be independent random variables and $f : \mathcal{X}^m \to \mathbb{R}$ a function such that changing any single variable X_i changes f by at most c_i:

$$\sup_{x,x'} |f(x_1, ..., x_i, ..., x_m) - f(x_1, ..., x_i', ..., x_m)| \le c_i \tag{43}$$

Then for any $\varepsilon > 0$:

$$\mathbb{P}[f(X_1, ..., X_m) - \mathbb{E}[f] \ge \varepsilon] \le \exp\left(\frac{-2\varepsilon^2}{\sum_{i=1}^m c_i^2}\right) \tag{44}$$

Use in learning theory: Applied to the supremum $\sup_{h\in\mathcal{H}} |R(h) - \hat{R}_S(h)|$, which changes by at most $O(\frac{1}{m})$ when one sample point changes. This gives the high-probability bound from the expectation bound.

Student Challenge

Exercise: Work through the full proof using the symmetrization lemma and Sauer's lemma. This is a key technique you'll use repeatedly.

🧩 **Synthesis Challenge**

Compare proof techniques: For $\mathcal{H}$ = linear classifiers in $\mathbb{R}^{10}$ with $\texttt{VCdim} = 11$ and $m = 1000$ samples:

1. Union bound over finite class (if discretized)
2. Union bound over growth function (Sauer's lemma)
3. Rademacher complexity (preview of Week 5)

What bound does each give? When is each most useful? How do they relate to the Fundamental Theorem?

Sample Complexity Bounds

The Fundamental Theorem and symmetrization give us **existence** results: they tell us learning is possible when $\mathrm{VCdim}(\mathcal{H}) < \infty$. But practitioners need concrete numbers: **How many samples do I actually need?**

This section provides the answer: we'll see both upper bounds ($O(\frac{d}{\varepsilon^2})$ samples suffice) and matching lower bounds ($\Omega(\frac{d}{\varepsilon^2})$ required in worst case). These tight bounds tell us that our theory is not just correct—it's **optimal**.

Upper Bounds

Theorem: Sample Complexity Upper Bound

Let $\mathcal{H}$ have $\mathrm{VCdim}(\mathcal{H}) = d < \infty$. For agnostic PAC learning with accuracy ε and confidence δ:

$$m = O\left(\frac{d\log\left(\frac{1}{\varepsilon}\right) + \log\left(\frac{1}{\delta}\right)}{\varepsilon^2}\right) \tag{45}$$

samples suffice.

This formalizes the sample complexity statement from the Fundamental Theorem (Section 5).

Proof: Follows from the Fundamental Theorem and uniform convergence bound.

Example: Concrete Sample Complexities

For $\varepsilon = 0.1$, $\delta = 0.05$:

- **Intervals on $\mathbb{R}$:** $d = 2$, so $m \approx 2000$
- **Linear classifiers in $\mathbb{R}^{10}$:** $d = 11$, so $m \approx 11000$
- **Linear classifiers in $\mathbb{R}^{100}$:** $d = 101$, so $m \approx 101000$
- **Axis-aligned rectangles in $\mathbb{R}^2$:** $d = 4$, so $m \approx 4000$

Notice the linear dependence on d!

Lower Bounds

Theorem: Sample Complexity Lower Bounds

Let $\mathcal{H}$ have $\text{VCdim}(\mathcal{H}) = d \geq 2$. In the realizable setting, some distributions require

$$m = \Omega\left(\frac{d}{\varepsilon} + \frac{\log(\frac{1}{\delta})}{\varepsilon}\right) \tag{46}$$

samples. In the agnostic setting, some distributions require

$$m = \Omega\left(\frac{d}{\varepsilon^2} + \frac{\log(\frac{1}{\delta})}{\varepsilon^2}\right) \tag{47}$$

samples.

Proof technique: Probabilistic method. Construct a hard distribution on a shattered set and show that with fewer samples, no algorithm can distinguish between random labelings.

Key Takeaway

The upper and lower bounds match up to logarithmic factors! This means our bounds are **tight**—we cannot do fundamentally better.

✓ **Checkpoint**

Quick check: Why is the agnostic lower bound $\Omega(\frac{d}{\varepsilon^2})$ while realizable is $\Omega(\frac{d}{\varepsilon})$?

Answer: In realizable, we know some $h \in \mathcal{H}$ has zero error. In agnostic, we must estimate error rates from noisy samples, requiring more data. The $\frac{1}{\varepsilon^2}$ vs. $\frac{1}{\varepsilon}$ reflects estimating a mean (agnostic) vs. testing if mean is zero (realizable).

🔨 **Construct Your Own**

Design a hard instance: For $\mathcal{H}$ = intervals on $\mathbb{R}$ with `VCdim` $= 2$, construct a distribution requiring $\Omega(\frac{2}{\varepsilon^2})$ samples in agnostic case.

Hint: Place probability mass on 3 points, make optimal classifier have error $\frac{1}{3}$. Any classifier from few samples will have error $\gg \frac{1}{3}$.

The No Free Lunch Theorem

So far we've assumed $\text{VCdim}(\mathcal{H}) < \infty$. But what if this doesn't hold?

The No Free Lunch Theorem delivers a stark answer: **No**. Without restricting the hypothesis class, learning is fundamentally impossible. This completes the picture: finite VC dimension isn't just sufficient for learning—it's also **necessary**.

Theorem: No Free Lunch Theorem (Informal)

For any algorithm A and sample size m, one can construct a distribution where A has constant probability of constant error. No learning algorithm works for all possible hypothesis classes.

Key insight: If $\text{VCdim}(\mathcal{H}) = \infty$, an adversary can construct scenarios where the learner sees only half the points, and labels on unseen points are independent of what was observed. This makes learning impossible.

Connection

The theorem says: you **must** restrict your hypothesis class (e.g., finite VC dimension) to make learning possible. Inductive bias is necessary, not optional.

Structural Risk Minimization (Brief Preview)

We've learned when learning is possible (finite VC dimension) and how many samples we need. But how do we choose among multiple hypothesis classes?

Structural Risk Minimization (SRM) provides a theoretical answer: choose the class minimizing training error **plus** a complexity penalty proportional to $\sqrt{\frac{d}{m}}$.

Definition: Structural Risk Minimization

Given nested classes $\mathcal{H}_1 \subseteq \mathcal{H}_2 \subseteq \ldots$, SRM chooses:

$$\hat{k} = \arg\min_k \left(\hat{R}_S(h_k) + \sqrt{\frac{\text{VCdim}(\mathcal{H}_k) \log m}{m}} \right) \tag{48}$$

where h_k minimizes training error in $\mathcal{H}_k$.

Interpretation: SRM balances fit (low training error) against complexity (VC dimension). This is the bias-variance tradeoff in action—the theoretical foundation for cross-validation and regularization (Week 5).

Connection

To Week 5: SRM motivates practical model selection via cross-validation and regularization penalties.

Summary and Connections

Key Insights

1. **Agnostic PAC learning** removes the realizability assumption and requires competing with the best hypothesis in $\mathcal{H}$
2. **Uniform convergence** is the key property: empirical risk must approximate true risk uniformly for all hypotheses
3. **The Fundamental Theorem** completely characterizes learnability: finite VC dimension is necessary and sufficient
4. **Symmetrization** is a powerful technique for deriving generalization bounds using ghost samples
5. **No Free Lunch** tells us that restrictions on $\mathcal{H}$ are necessary—learning requires inductive bias
6. **Sample complexity** scales as $O\left(\frac{d}{\varepsilon^2}\right)$ where $d = \text{VCdim}(\mathcal{H})$, and this is tight

Key Takeaway

The Fundamental Theorem of Statistical Learning is the cornerstone of learning theory. It tells us exactly when learning is possible and how many samples we need. Everything else is refinement and application of this core result.

📋 Proof Template: Proving uniform convergence for your class

To show $\mathcal{H}$ satisfies uniform convergence:

1. **Verify finite VC:** Show $\text{VCdim}(\mathcal{H}) = d < \infty$
2. **Apply Sauer:** $\Pi_{\mathcal{H}}(m) \leq \left(\frac{em}{d}\right)^d$
3. **Symmetrization:** Introduce ghost sample, bound $\mathbb{P}\left[\sup_{h\in\mathcal{H}}|R(h) - \hat{R}_S(\mathsf{h})| > \varepsilon\right]$
4. **Union bound:** Over $\Pi_{\mathcal{H}}(2m)$ behaviors
5. **Concentration:** Apply Hoeffding to each
6. **Solve for m:** Get $m = O\left(\frac{d\log(\frac{1}{\varepsilon}) + \log(\frac{1}{\delta})}{\varepsilon^2}\right)$

📋 Proof Template: Proving sample complexity lower bound

To show $\mathcal{H}$ requires $\Omega\left(\frac{d}{\varepsilon^2}\right)$ samples:

1. **Find shattered set:** Identify set C of size d that $\mathcal{H}$ shatters
2. **Construct distribution:** Uniform over C
3. **Random labeling:** Choose target uniformly
4. **Analyze learning:** With $m < c\frac{d}{\varepsilon^2}$, learner sees too few points
5. **Calculate error:** Expected error 0.5 on unseen; cannot achieve $\leq \varepsilon$

Connections to Other Topics

Connection

To Week 4: Bias-variance tradeoff and regularization implement the complexity-fit balance in practice.

To Week 6-7: Boosting and ensemble methods—how does combining weak learners affect complexity?

To Week 11: Deep learning theory. Why do neural networks generalize despite having huge VC dimension?

Connection

Building on Week 2: Last week we computed VC dimensions for specific classes (intervals, rectangles, linear classifiers). This week we saw **why** VC dimension matters: it's the key quantity controlling sample complexity. The classes we analyzed are all learnable precisely because they have finite VC dimension.

Connection

Looking ahead to Week 4: Next week focuses on **practical** learning:

- Cross-validation for model selection (empirical alternative to SRM)
- Regularization (penalizing complexity in objective)
- Bias-variance decomposition (quantifying approximation-estimation trade-off)
- Validation set methods (practical tools for avoiding overfitting)

🧠 **Mental Model: The Learning Theory Landscape**

The chain of learning theory:

Hypothesis Class $\mathcal{H}$ $\to$ Complexity $\mathrm{VCdim}(\mathcal{H}) = d$ $\to$ Sample Complexity $m = O(\frac{d}{\varepsilon^2})$ $\to$ Uniform Convergence $\sup_{h\in\mathcal{H}} |R(h) - \hat{R}_S(h)| \leq \varepsilon$ $\to$ ERM Success $R(h_S) \leq \min_{h\in\mathcal{H}} R(h) + \varepsilon$

Every arrow is an implication in the Fundamental Theorem. The chain holds iff $d < \infty$.

Closing Challenge

Opening Challenge

Revisiting the Opening Question:

Your training data is noisy and your hypothesis class may be misspecified. Can you still learn?

Answer: Yes! The Fundamental Theorem guarantees that if $\mathrm{VCdim}(\mathcal{H}) < \infty$, then:

- ERM will find a hypothesis close to the best in $\mathcal{H}$
- Sample complexity is $O(\frac{d}{\varepsilon^2})$
- This holds regardless of noise or misspecification

Practical lesson:

- Choose hypothesis classes with finite (and small) VC dimension
- Use enough samples: $m \gg \frac{d}{\varepsilon^2}$
- Balance model complexity with sample size
- Validate on held-out data to estimate true risk

Author's Take

The Fundamental Theorem provides a complete answer within the standard binary classification framework. It connects four seemingly different concepts (uniform convergence, PAC learnability, VC dimension) into a single unified framework.

In practice, most modern machine learning operates in the agnostic setting—we rarely have perfect data or perfectly specified models. The Fundamental Theorem tells us this is okay: as long as our hypothesis class has finite VC dimension and we have enough samples, learning is possible.

The key insight is that **complexity matters**. Not all hypothesis classes are learnable, and the sample complexity depends critically on the VC dimension. This motivates the entire field of model selection and regularization, which we'll explore in coming weeks.

🧩 Synthesis Challenge

Capstone: Design a complete learning system

Problem: Predict disease from 20 biomarkers with $m = 500$ samples.

Tasks:

1. Propose three hypothesis classes and estimate their VC dimensions
2. Which are learnable with $\varepsilon = 0.1$, $\delta = 0.05$?
3. Use SRM with training errors $\texttt{Remp}_1 = 0.15$, $\texttt{Remp}_2 = 0.10$, $\texttt{Remp}_3 = 0.05$
4. How much more data to make the complex class competitive?
5. What inductive biases are implicit in your choices?

This integrates: agnostic PAC, VC dimension, sample complexity, uniform convergence, SRM, No Free Lunch.

Example: Medical diagnosis finale

You started with noisy data and misspecified model (linear for nonlinear problem). Through Week 3′s theory:

- Your problem is **agnostic** (best linear has 15% error)
- With $m = 1000$ and `VCdim` $= 11$, you can find classifier with 17% error (Fundamental Theorem)
- ERM works because uniform convergence holds (finite VC)
- You needed $m = O(\frac{11}{0.01}) \approx 1100$ samples ✓
- Alternative classes (quadratic) might fit better but need more samples or risk overfitting (SRM)

This is learning theory in action: rigorous guarantees for messy, real-world problems.